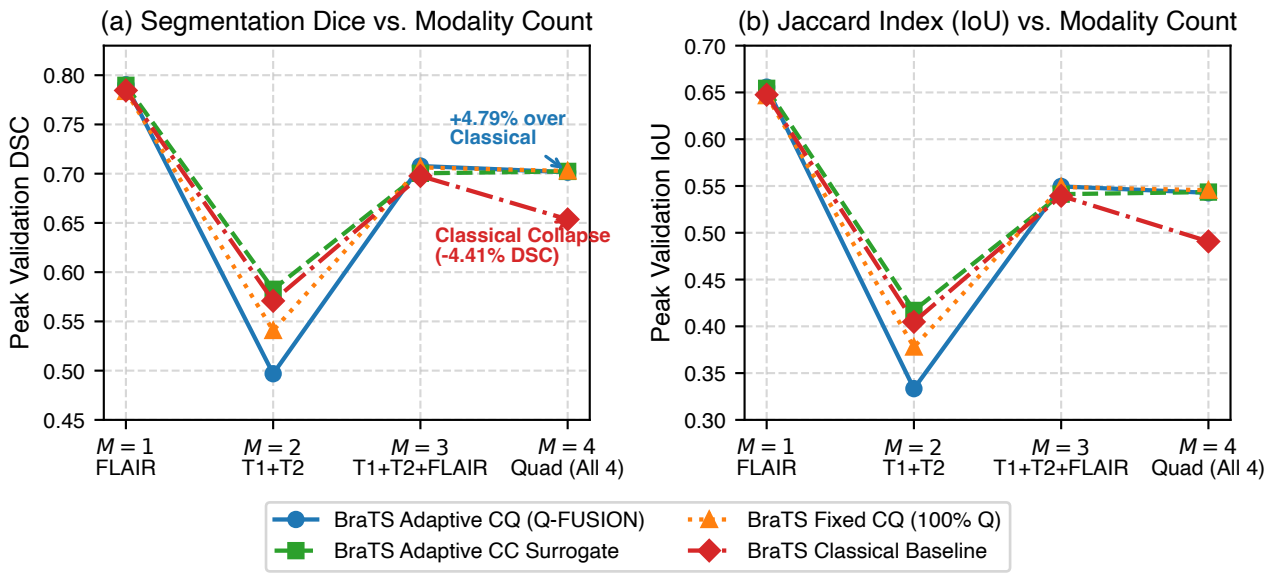
(a) Segmentation Dice vs. Modality Count
Peak Validation DSC
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0.45
+4.79% over Classical
Classical Collapse (-4.41% DSC)
M = 1 FLAIR
M = 2 T1+T2
M = 3 T1+T2+FLAIR
M = 4 Quad (All 4)
(b) Jaccard Index (IoU) vs. Modality Count
Peak Validation IoU
0.70
0.65
0.60
0.55
0.50
0.45
0.40
0.35
0.30
M = 1 FLAIR
M = 2 T1+T2
M = 3 T1+T2+FLAIR
M = 4 Quad (All 4)
BraTS Adaptive CQ (Q-FUSION)
BraTS Adaptive CC Surrogate
BraTS Fixed CQ (100% Q)
BraTS Classical Baseline